

Norway

SLEIPNER

Volve F

F-15

F-15B

Design: F-15B

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Site:	Volve F	MD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Well:	F-15	North Reference:	Grid
Wellbore:	F-15B	Survey Calculation Method:	Minimum Curvature
Design:	F-15B	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude: 58° 26' 29.8068 N
From: Map		Easting:	435,050.02 m	Longitude: 1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence: -0.95 °

Well	F-15			
Well Position	+N/-S	-3.17 m	Northing:	6,478,560.36 m
	+E/-W	3.53 m	Easting:	435,053.55 m
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m
			Water Depth:	91.00 m

Wellbore	F-15B				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	BGGM2008	31.12.2008	-2.06	71.62	50,396

Design	F-15B			
Audit Notes:				
Version:	1.0	Phase:	ACTUAL	Tie On Depth: 3,210.00
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)
	145.90	-3.17	3.53	226.62

Survey Program	Date	19.07.2010		
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description
153.40	1,364.50	12 1/4" F-15 Drop Gyro (F-15A)	Wellbore Surveyor, stat	Gyro Tool from GD
1,383.00	2,513.00	12 1/4" Drop Gyro F-15A Revised (F-15A)	Wellbore Surveyor, stat	Gyro Tool from GD
2,544.30	3,149.35	8 1/2" Drop gyro F-15A (F-15A)	Wellbore Surveyor, stat	Gyro Tool from GD
3,149.35	3,476.85	8 1/2" MWD F15B (F-15B)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	-3.17	3.53	0.00	0.000	0.000	0.000	
153.40	0.15	253.69	153.40	-3.17	3.52	0.01	0.600	0.600	0.000	
234.20	0.28	146.02	234.20	-3.37	3.53	0.14	0.132	0.048	-39.976	
274.60	0.78	136.21	274.60	-3.65	3.78	0.15	0.376	0.371	-7.285	
315.00	1.46	128.29	314.99	-4.16	4.37	0.07	0.517	0.505	-5.881	
355.30	3.41	124.07	355.25	-5.15	5.77	-0.26	1.457	1.452	-3.141	
395.70	5.13	126.98	395.54	-6.91	8.21	-0.83	1.287	1.277	2.161	
436.10	5.52	139.07	435.76	-9.47	10.92	-1.05	0.880	0.290	8.978	

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Wellbore:	F-15B	Survey Calculation Method:	Minimum Curvature
Design:	F-15B	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
474.40	5.62	152.33	473.89	-12.52	13.00	-0.46	1.009	0.078	10.386
514.80	6.19	163.05	514.07	-16.36	14.55	1.05	0.920	0.423	7.960
557.10	7.03	171.79	556.09	-21.10	15.59	3.55	0.928	0.596	6.199
597.50	7.74	174.05	596.16	-26.25	16.22	6.63	0.570	0.527	1.678
637.80	8.49	174.54	636.05	-31.91	16.79	10.11	0.561	0.558	0.365
678.20	8.94	175.57	675.99	-38.01	17.31	13.91	0.354	0.334	0.765
718.50	9.34	177.45	715.77	-44.40	17.70	18.02	0.372	0.298	1.400
758.90	10.22	185.51	755.59	-51.24	17.50	22.86	1.207	0.653	5.985
799.30	11.89	193.59	795.24	-58.86	16.18	29.05	1.689	1.240	6.000
839.70	13.68	199.13	834.64	-67.42	13.64	36.78	1.610	1.329	4.114
880.00	15.14	201.54	873.67	-76.81	10.14	45.78	1.175	1.087	1.794
920.40	13.72	203.30	912.79	-86.12	6.31	54.95	1.103	-1.054	1.307
960.80	14.53	201.31	951.97	-95.24	2.57	63.93	0.701	0.601	-1.478
1,001.10	16.51	200.62	990.80	-105.31	-1.28	73.65	1.480	1.474	-0.514
1,041.60	16.33	201.19	1,029.65	-116.01	-5.37	83.97	0.179	-0.133	0.422
1,081.90	16.18	205.25	1,068.34	-126.37	-9.81	94.31	0.853	-0.112	3.022
1,122.30	17.54	211.88	1,107.01	-136.63	-15.43	105.44	1.747	1.010	4.923
1,162.60	17.66	221.21	1,145.43	-146.39	-22.66	117.40	2.100	0.089	6.945
1,203.00	16.05	230.28	1,184.10	-154.57	-31.00	129.08	2.286	-1.196	6.735
1,243.30	15.33	232.77	1,222.90	-161.35	-39.52	139.94	0.734	-0.536	1.854
1,283.80	18.17	237.03	1,261.68	-168.03	-49.09	151.47	2.291	2.104	3.156
1,324.10	21.21	241.08	1,299.62	-174.98	-60.74	164.72	2.479	2.263	3.015
1,364.50	24.71	243.03	1,336.81	-182.34	-74.67	179.90	2.659	2.599	1.448
1,383.00	26.95	242.95	1,353.46	-186.01	-81.85	187.63	3.633	3.632	-0.130
1,406.00	25.71	234.70	1,374.08	-191.26	-90.57	197.58	5.035	-1.617	-10.761
1,445.00	24.44	222.89	1,409.43	-202.07	-102.97	214.01	3.966	-0.977	-9.085
1,487.00	21.59	210.72	1,448.10	-215.09	-112.84	230.13	3.952	-2.036	-8.693
1,544.00	20.56	192.78	1,501.35	-233.88	-120.42	248.55	3.425	-0.542	-9.442
1,584.00	21.08	181.10	1,538.75	-247.93	-122.11	259.42	3.133	0.390	-8.760
1,625.00	21.07	170.97	1,577.02	-262.59	-121.10	268.75	2.662	-0.007	-7.412
1,665.00	20.92	159.85	1,614.38	-276.40	-117.51	275.63	2.986	-0.112	-8.340
1,705.00	22.16	151.36	1,651.59	-289.73	-111.43	280.36	2.513	0.930	-6.367
1,746.00	24.79	147.70	1,689.20	-303.78	-103.13	283.98	2.199	1.924	-2.678
1,786.00	27.24	144.45	1,725.15	-318.32	-93.32	286.84	2.125	1.837	-2.437
1,827.00	27.70	143.46	1,761.52	-333.61	-82.19	289.25	0.474	0.337	-0.724
1,867.00	28.14	143.47	1,796.87	-348.66	-71.04	291.49	0.330	0.330	0.007
1,907.00	27.08	143.10	1,832.31	-363.52	-59.96	293.64	0.805	-0.795	-0.277
1,948.00	27.31	143.53	1,868.78	-378.55	-48.77	295.82	0.221	0.168	0.315
1,988.00	27.71	143.72	1,904.26	-393.42	-37.81	298.07	0.307	0.300	0.142
2,028.00	27.86	145.00	1,939.64	-408.58	-26.95	300.58	0.461	0.112	0.960
2,069.00	27.42	149.82	1,975.97	-424.59	-16.70	304.14	1.667	-0.322	3.527
2,109.00	26.74	154.96	2,011.59	-440.70	-8.26	309.07	1.827	-0.510	3.855
2,149.00	25.09	162.89	2,047.58	-456.97	-1.96	315.66	2.876	-1.237	5.947

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Site:	Volve F	MD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Well:	F-15	North Reference:	Grid
Wellbore:	F-15B	Survey Calculation Method:	Minimum Curvature
Design:	F-15B	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
2,190.00	24.05	170.43	2,084.87	-473.52	1.99	324.15	2.415	-0.761	5.517	
2,230.00	24.40	174.59	2,121.35	-489.78	4.13	333.77	1.307	0.262	3.120	
2,270.00	23.87	181.76	2,157.87	-506.10	4.66	344.59	2.233	-0.397	5.377	
2,311.00	24.72	188.97	2,195.24	-522.86	3.06	357.27	2.256	0.622	5.276	
2,351.00	25.05	194.20	2,231.53	-539.34	-0.32	371.04	1.669	0.247	3.922	
2,392.00	25.95	202.77	2,268.55	-556.03	-5.92	386.57	2.776	0.659	6.271	
2,432.00	27.22	209.99	2,304.33	-572.03	-13.88	403.35	2.602	0.952	5.415	
2,486.00	28.73	219.32	2,352.04	-592.78	-28.29	428.07	2.569	0.839	5.183	
2,513.00	30.35	223.67	2,375.54	-602.73	-37.11	441.32	2.985	1.800	4.833	
2,544.30	32.21	227.05	2,402.29	-614.14	-48.68	457.56	2.450	1.783	3.240	
2,584.60	36.06	231.49	2,435.64	-628.85	-65.83	480.13	3.412	2.866	3.305	
2,624.91	39.18	235.66	2,467.57	-643.42	-85.64	504.54	2.996	2.322	3.103	
2,665.26	41.67	241.21	2,498.30	-657.08	-107.93	530.12	3.252	1.851	4.126	
2,705.63	43.76	244.74	2,527.96	-669.50	-132.32	556.38	2.361	1.553	2.623	
2,745.99	46.14	249.81	2,556.53	-680.49	-158.61	583.03	3.196	1.769	3.769	
2,786.36	48.79	250.79	2,583.82	-690.51	-186.62	610.27	2.041	1.969	0.728	
2,826.70	51.38	251.73	2,609.70	-700.44	-215.92	638.39	1.999	1.926	0.699	
2,867.05	51.74	251.02	2,634.79	-710.54	-245.86	667.09	0.493	0.268	-0.528	
2,907.38	49.19	251.75	2,660.46	-720.47	-275.34	695.33	1.943	-1.897	0.543	
2,947.73	49.52	257.15	2,686.75	-728.67	-304.81	722.39	3.056	0.245	4.015	
2,988.07	50.68	263.14	2,712.64	-733.95	-335.28	748.16	3.524	0.863	4.455	
3,028.40	52.73	265.32	2,737.64	-737.12	-366.77	773.23	1.986	1.525	1.622	
3,068.70	57.54	267.07	2,760.67	-739.30	-399.75	798.70	3.737	3.581	1.303	
3,109.03	59.86	268.74	2,781.62	-740.55	-434.18	824.58	2.026	1.726	1.242	
3,149.35	60.21	270.16	2,801.76	-740.89	-469.11	850.20	0.952	0.260	1.057	
3,158.94	59.52	269.88	2,806.57	-740.88	-477.40	856.23	2.288	-2.158	-0.876	
3,194.82	59.29	272.22	2,824.84	-740.32	-508.28	878.28	1.695	-0.192	1.957	
3,234.92	54.92	274.74	2,846.61	-738.29	-541.88	901.31	3.632	-3.269	1.885	
3,274.52	53.75	278.21	2,869.71	-734.67	-573.84	922.06	2.312	-0.886	2.629	
3,315.03	51.13	279.70	2,894.40	-729.68	-605.56	941.68	2.128	-1.940	1.103	
3,355.78	50.70	280.07	2,920.09	-724.25	-636.72	960.60	0.381	-0.317	0.272	
3,396.46	48.09	281.46	2,946.57	-718.49	-667.06	978.70	2.076	-1.925	1.025	
3,436.37	46.32	282.43	2,973.68	-712.43	-695.71	995.36	1.434	-1.330	0.729	
3,476.85	44.43	281.71	3,002.11	-706.41	-723.88	1,011.70	1.451	-1.401	-0.534	
3,497.00	44.43	281.71	3,016.50	-703.54	-737.69	1,019.77	0.000	0.000	0.000	

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Project:	SLEIPNER	TVD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Site:	Volve F	MD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Well:	F-15	North Reference:	Grid
Wellbore:	F-15B	Survey Calculation Method:	Minimum Curvature
Design:	F-15B	Database:	Production EDM P246N

Design Targets									
Target Name	Dip Angle	Dip Dir.	TVD	+N/-S	+E/-W	Northing	Easting	Latitude	Longitude
- hit/miss target	(°)	(°)	(m)	(m)	(m)	(m)	(m)		
- Shape									
F15A Base Hugin Tgt 1 (	0.00	0.00	3,017.90	-702.77	-741.28	6,477,861.00	434,309.00	58° 26' 6.6987 N	1° 52' 29.9690 E
- actual wellpath misses target center by 3.93m at 3497.00m MD (3016.50 TVD, -703.54 N, -737.69 E)									
- Rectangle (sides W27.36 H37.84 D0.00)									

Casing Points					
Measured Depth	Vertical Depth			Casing Diameter	Hole Diameter
(m)	(m)		Name	(in)	(in)
220.60	220.60	30"		30.000	36.000
1,368.40	1,340.35	20"		20.000	26.000
2,561.80	2,416.96	13 3/8"		13.375	16.000

Checked By: _____ Approved By: _____ Date: _____